Giving Life to Private (Rated) Credit

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- **Our Setting:** The U.S. life insurance industry (**$500B privately rated bonds**)

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    - High yield $\sim$2 notches | Investment grade $\sim 1/3$ notch
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  - **Answer: Yes.**
    - Investment grade: ∼1pp ↑, ∼ 7% ratings gap
    - High yield (bonds): ∼2.7pp ↑, ∼ 66% ratings gap

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- **Q3: Does the demand for private ratings reach private bond issuance?**
  - **Answer: Partly.**
  - Firms connected to exposed insurers ↑ private (144A) IG issuance
  - Concentrated in *foreign-domiciled* issuers

Literature

- **Credit Ratings in Opaque Markets**

  Post-GFC: Skreta & Veldkamp 2009, Benmelech & Dlugosz 2010, Bolton et al 2012, Griffin & Tang 2012

  Ratings Agencies: Opp et al 2013, Dimitrov et al 2015, Bruno et al 2016, Behr et al 2018

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- **Regulation, Insurer Asset Demand, and Private Markets**

  Private Markets: Kirti & Sarin 2024, Meisenzahl et al 2025, Carlino et al 2025, Huber et al 2026

  Portfolio Choice: Ellul et al 2011, Ge & Weisbach 2021, Becker et al 2022, Bhardwaj et al 2025, Li 2026

  Valuation: Li, Oh, & Ricciardi 2026, Sen, Sharma, & Viceira 2026, Drexler & Polacek 2026

  Regulatory Arbitrage: Becker & Ivashina 2015, Merrill et al 2019, Fringuellotti & Santos 2021, Weinlich 2024

  This paper: both channels in one setting, split by grade and by designation route

  cap'l requirements → demand for priv. ratings → inflation (HY) vs. access (IG) → issuance

Institutional Details

Risk-Based Capital Timeline

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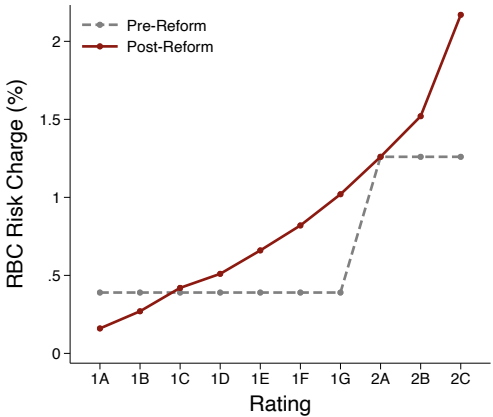
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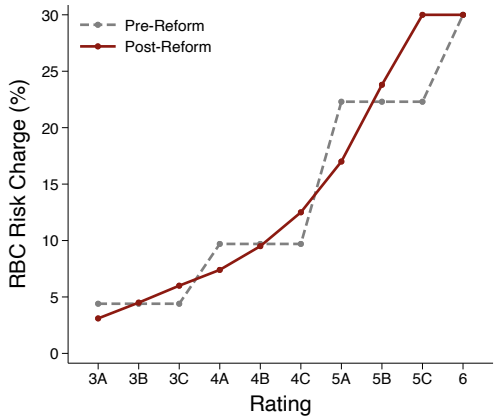
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  2021 (Jun): NAIC adopts proposal, implements with 2021Q4 filings

2021 Risk-Based Capital Reform



(a) Investment Grade (NAIC 1-2)



(b) High Yield (NAIC 3-6)

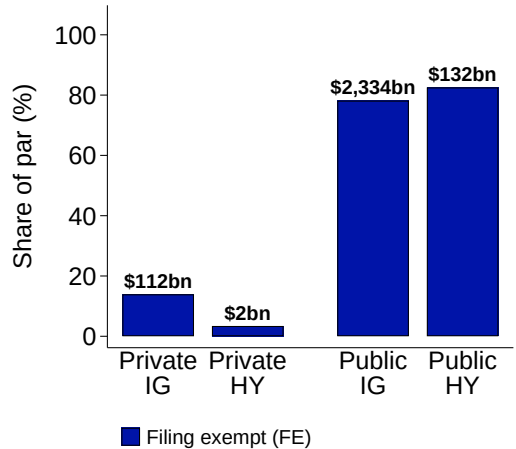
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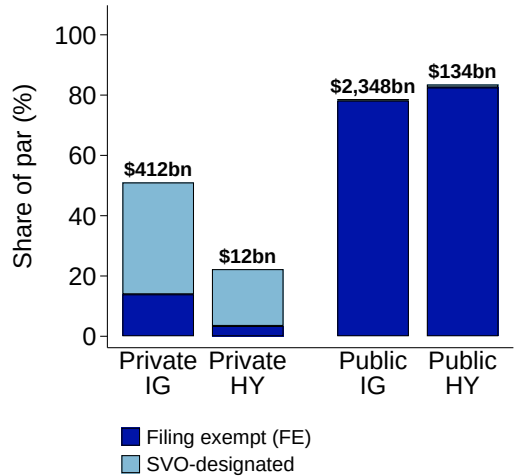


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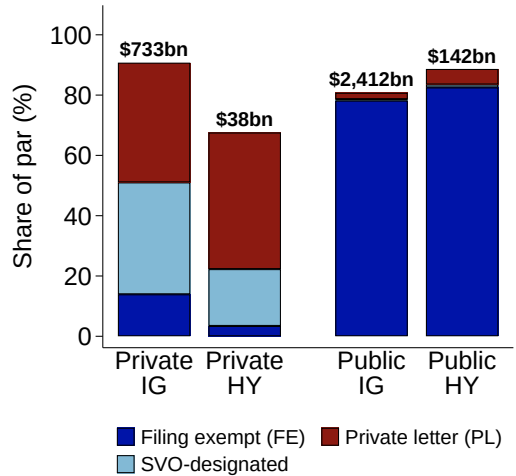
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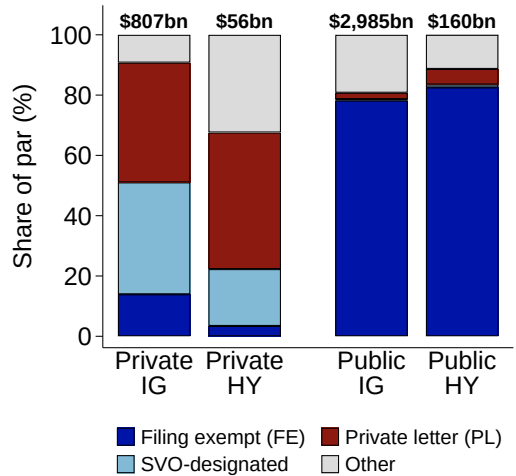
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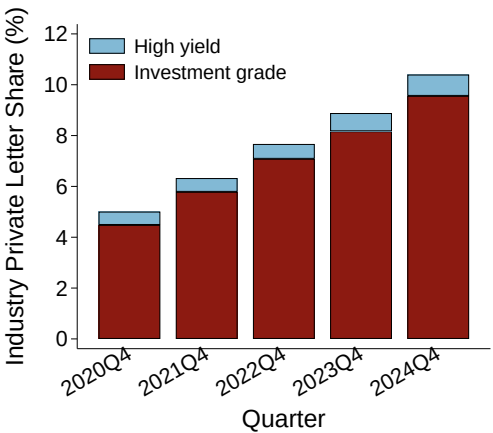


Data

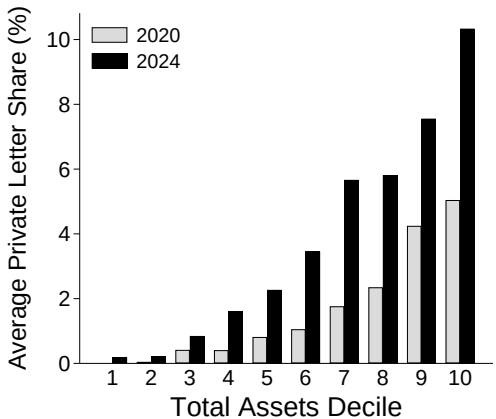
- **NAIC Schedule D** — insurer × bond holdings panel, 2020-2024
  - Key vars: par value, carrying value, yields, risk factors, SVO Symbol (PL)
  - Extras: callability, capital structure, maturity, private placement (PP), asset type
- **Mergent FISD & Refinitiv** — firm-level bond issuance panel, 2018-2024
  - Private placement (PP) — typically "no SEC filing" or 144A
  - Public credit ratings
- **WRDS Bond Return** — public bond market yield panel
  - used for robustness only

Characteristics of Bonds with Private Letters

Fact 1: Private Letter Holdings Doubled in 5 Years

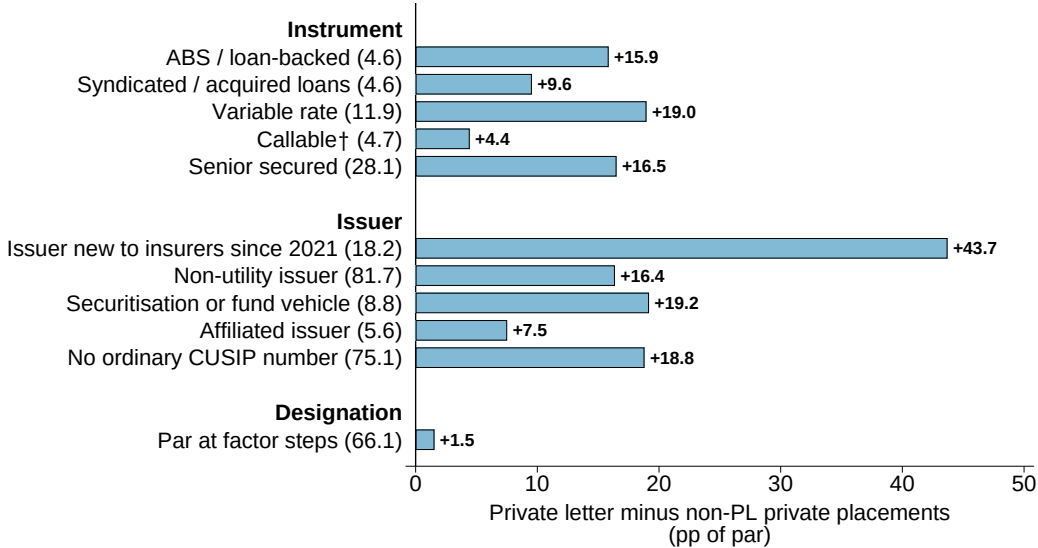


(a) PL share of industry par



(b) Growth by size, 2020 → 2024

Fact 2: Bonds with Private Ratings are Different

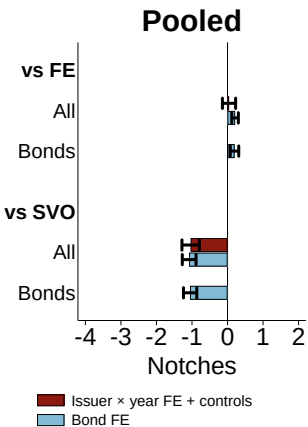


Fact 3: Bonds with Private Ratings have Cheaper Capital Charges

$$\text{(Rating Notch)}_{ibt} = \beta \textbf{ (Private Rating)}_{it} + \text{controls} + \text{fixed effects} + \varepsilon_{ibt}$$

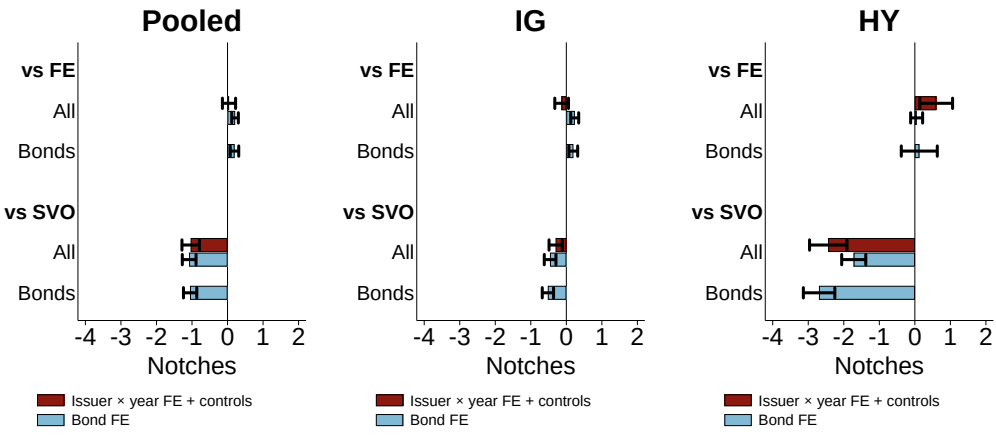
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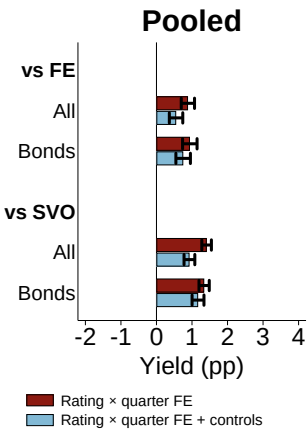


Fact 4: Bonds with Private Ratings have Capital Efficient Returns

$$\text{Yield}_{ibt} = \beta \textbf{ (Private Rating)}_{it} + \text{controls} + \text{rating} \times \text{quarter FE} + \varepsilon_{ibt}$$

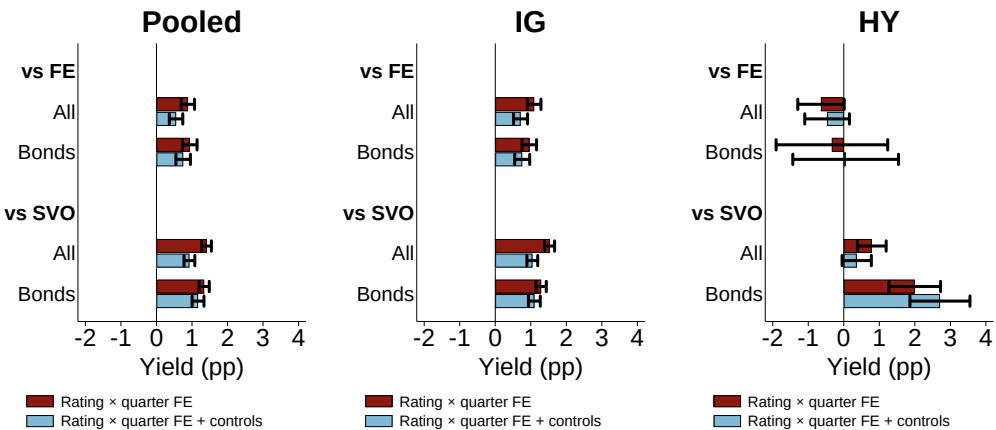
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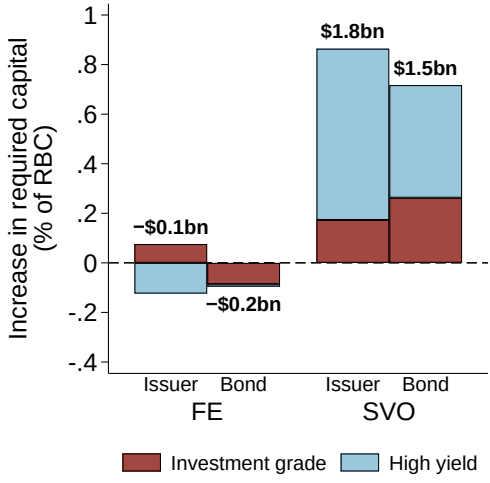


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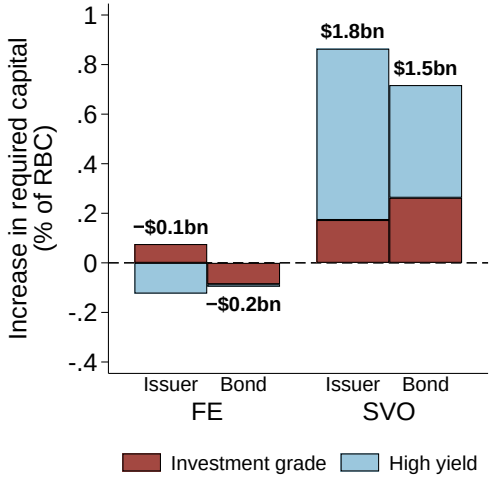


What are the consequences of ratings inflation?

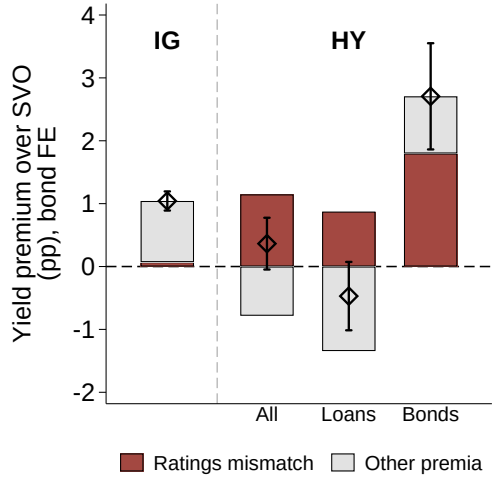


(a) Required Capital

What are the consequences of ratings inflation?



(a) Required Capital



(b) Yield Decomposition

Summary of the Facts

Bonds with private ratings ...

(Fact 1) ... have doubled their portfolio share on insurers' balance sheets in 5 years

(Fact 2) ... are more likely structured, new, and private-credit-like

(Fact 3) ... appear inflated, but mostly in the high yield segment

(Fact 4) ... provide capital-efficient yields relative to similarly-rated non-PL bonds

(C'fls) ... reduce required capital modestly and drive yield gaps for HY bonds

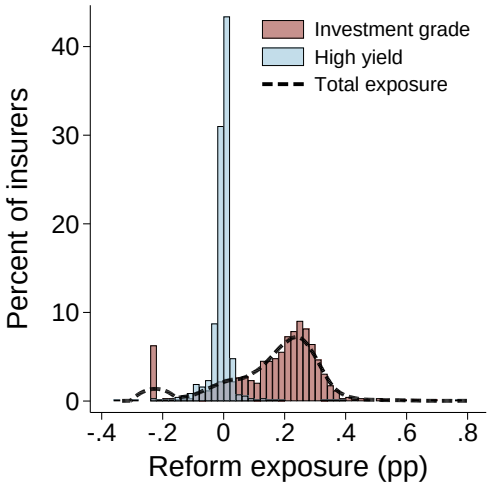
Do insurers use private ratings for capital relief?

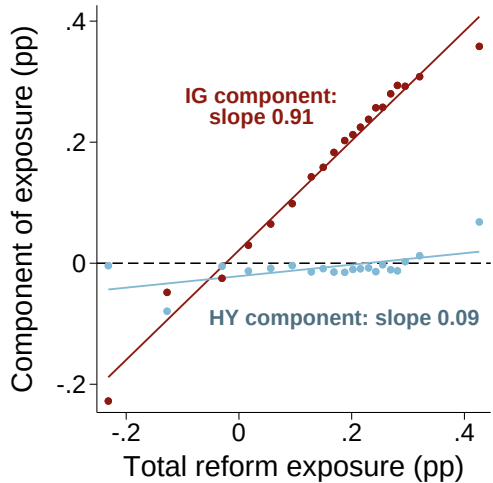
The RBC Reform – an Exogenous Shock to Required Capital

- **RBC Reform:** exogenous *level* changes in risk-charges *across ratings*
- **Assumption:** **Δ(risk factor)** *exogenous* to **pre-reform portfolio weights**
  - Timing: new risk factors announced in June 2021 vs. portfolio weights in 2020Q4
  - Levels: variations across ratings that receive the same factor pre-reform
- **Reform Exposure:**

$$\text{Exposure}_j = \sum_c (\text{bond portfolio weight})_{\boldsymbol{jc,2020}} \times \Delta(\text{risk-factor})_c$$

Reform Exposure: (Almost Entirely) an Investment-Grade Shock



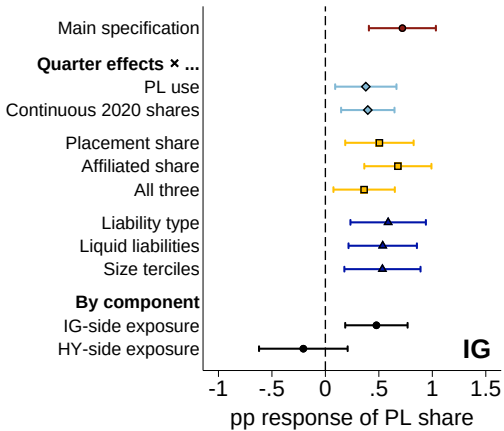


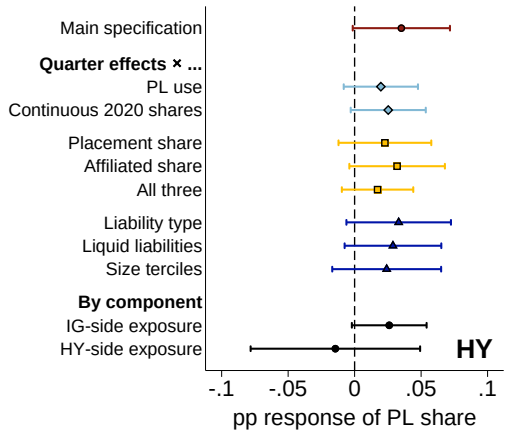
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$$\text{PL share}_{jt} = \beta\, \text{Exposure}_j \times \text{Post}_t + \text{controls} + \alpha_t + \delta_j + \varepsilon_{jt} \quad \text{(per s.d. of exposure)}$$

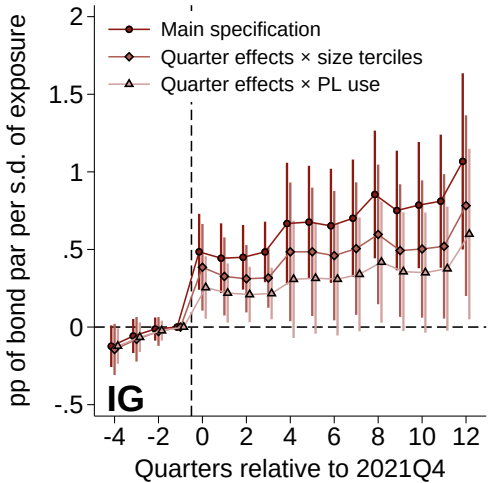
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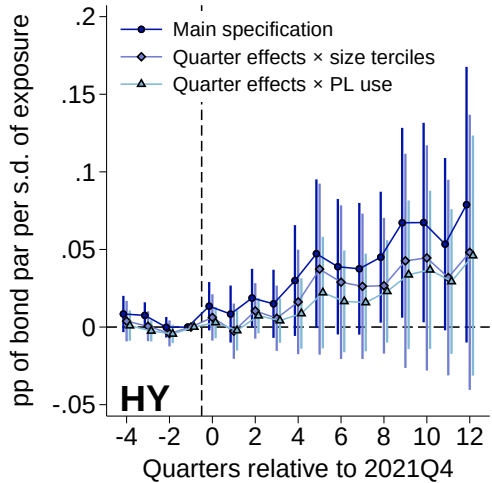
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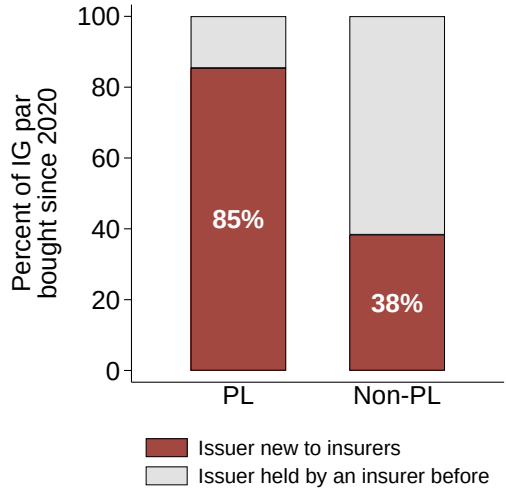
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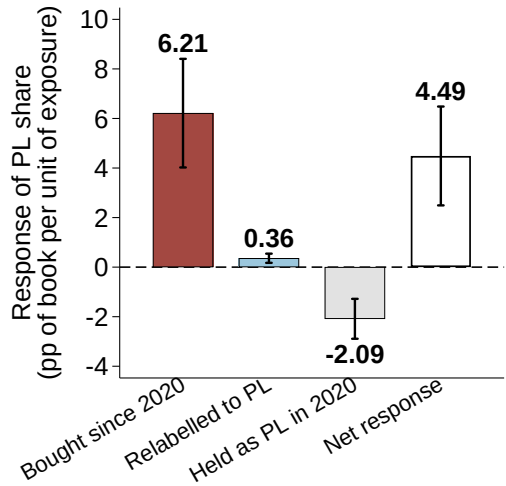
What did they buy?

- Most new PL bonds are also new issuers wrt the industry



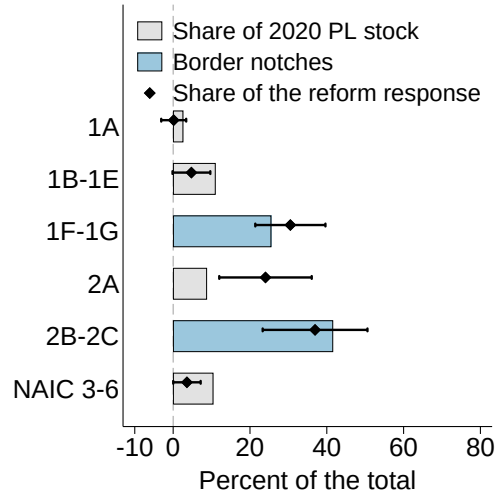
What did they buy?

- Most new PL bonds are also new issuers wrt the industry
- Reform response concentrated in **new issuers**, not **relabels**

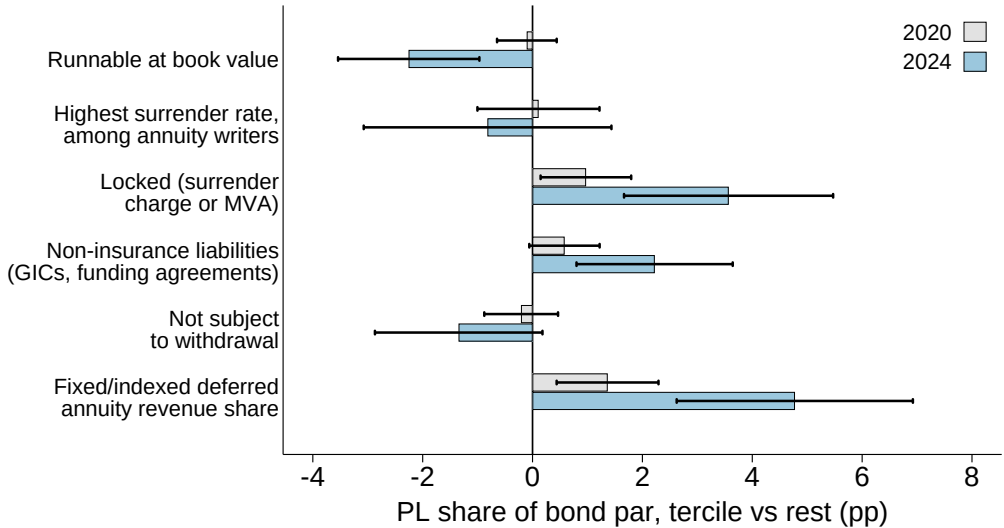


What did they buy?

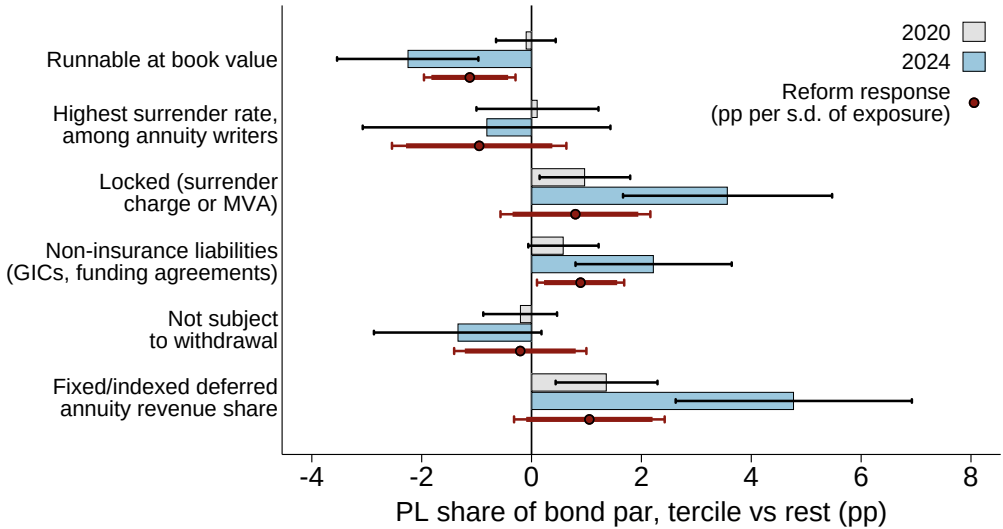
- Most new PL bonds are also new issuers wrt the industry
- Reform response concentrated in **new issuers**, not **relabels**
- No evidence of reform response **bunching** at most affected notches



Who Buys It? The Role of Liability Structure



Who Buys It? The Role of Liability Structure



Does Private Ratings Demand $\rightarrow$ Private Debt Issuance?

Did Regulation Shift Private Issuance Incentives?

- **Private ratings** and **placements** not observed at the issuance/firm level
  - We only see private ratings on insurers' balance sheets
  - The same bond can carry different ratings across different investors

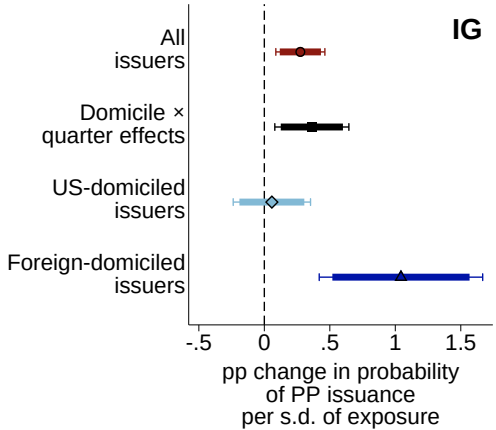
- Look at **144A issuance** instead
  - Among newly issued bonds, 144A bonds are much more likely to bear PL (in paper)
  - Differences in domestic vs. foreign issuance ($\sim 1/6$ of insurers' balance sheets)

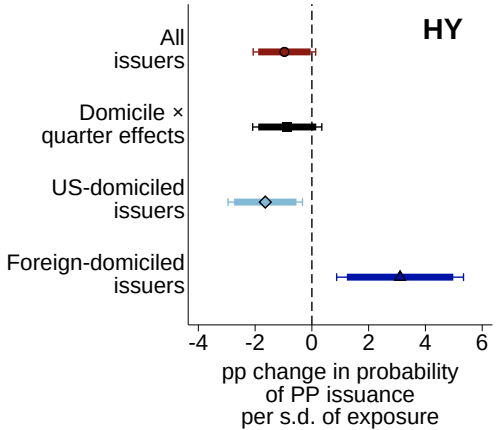
- **Firm-level reform exposure:**

$$\text{Firm Exposure}_i = \sum_{j=1}^{J} \left(\frac{\text{Firm } i\text{'s Bonds Held by Insurer } j}{\text{Firm } i\text{'s Total Bond Outstanding}} \right) \times \text{Insurer Exposure}_j$$

Did Regulation Shift Private Issuance Incentives?

$$(\text{Firm } i\text{'s } \textbf{144A} \text{ Issuance})_t = \beta(\text{Firm Exposure})_i \times \text{Post}_t + \alpha_i + \delta_t + \varepsilon_{it}$$





▸ New 144A → PL

Conclusion

Private Ratings Buy Capital Relief: Inflation in HY, Access in IG

- **PLRs are a capital-efficient route into private debt; arbitrage is a HY story**
  - Inflation vs. the SVO is HY; the ∼90bp IG premium is not a rating artefact
  - Tighter charges → **IG** private ratings via new issuers, esp. stable liabilities
  - Ratings demand reaches 144A IG issuance, mostly foreign-domiciled firms

- **What's next?**
  - **Efficiency:** a model of capital-constrained insurers with a private-rating technology
  - **Premium:** illiquidity, origination rents, or credit risk the designation misses?
  - **Issuance:** who are the foreign issuers, and what did they substitute from?

Thank you!

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Appendix

Likelihood of Newly Acquired 144A's Receiving PLR

$$\text{AnyPL}_{bt} = \beta\ 144\text{A}_b + \alpha_t + \varepsilon_{bt}$$

	0 quarters	$\leq$ 4 quarters	$\leq$ 8 quarters
Privately Placed	1.78***	1.61***	1.46***
	(0.40)	(0.38)	(0.39)
% Rel. to non-PP	493%	400%	331%
Quarter FE	✓	✓	✓
Observations	8,371	5,874	4,012
BIC	596.3	686.9	591.7
Pseudo R^2	0.09	0.08	0.05

▸ Back